# DNS SERVER PRACTICAL ASSIGNMENT

BIND on Linux • DNS Zone • A Record • CNAME • Troubleshooting

**Student Name:** _____ **Class:** _____
**Date:** _____ **Subject:** Networking

**Platform:** Ubuntu Linux VM with BIND9. This is a local lab exercise. The address 192.168.1.50 is the required example record value; it does not prove that a web server is running there.

## 1. Install BIND and create the musicmania.in DNS zone

1. Open Terminal in your Ubuntu VM and update the package list:

```
sudo apt update
```

2. Install BIND9 and DNS tools:

```
sudo apt install bind9 bind9-utils dnsutils -y
```

3. Open the local BIND configuration:

```
sudo nano /etc/bind/named.conf.local
```

Add this zone declaration and save the file:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

4. Create the zone file:

```
sudo cp /etc/bind/db.local /etc/bind/db.musicmania.in
sudo nano /etc/bind/db.musicmania.in
```

5. Replace its contents with the following zone data:

```
$TTL    604800
@       IN      SOA     ns1.musicmania.in. admin.musicmania.in. (
                        2026111101 ; Serial
                        604800     ; Refresh
                        86400      ; Retry
                        2419200    ; Expire
                        604800 )   ; Negative Cache TTL
@       IN      NS      ns1.musicmania.in.
ns1     IN      A       192.168.1.50
www     IN      A       192.168.1.50
play    IN      CNAME   www.musicmania.in.
```

6. Validate the configuration and zone file:

```
sudo named-checkconf
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

A successful zone check should say that the zone loaded successfully. Correct any reported errors before continuing.

## 2. Verify the A record for www.musicmania.in

The A record maps the name www.musicmania.in to the IPv4 address 192.168.1.50. Restart BIND and check its status:

```
sudo systemctl restart bind9
sudo systemctl status bind9
```

Query the local DNS server directly:

```
nslookup www.musicmania.in 127.0.0.1
# or
dig @127.0.0.1 www.musicmania.in A
```

**Expected answer:** www.musicmania.in should resolve to 192.168.1.50. Write your actual result below.

| Test | Your result |
|---|---|
| A record lookup | _____ |

| | |
|---|---|
| Screenshot / notes | _____<br/>_____ |

## 3. Add and test the CNAME record for play.musicmania.in

A CNAME record makes one DNS name an alias of another DNS name. The zone file should contain:

```
play    IN      CNAME   www.musicmania.in.
```

After editing the zone file, increase the SOA serial number, validate it, and reload the zone:

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo rndc reload musicmania.in
```

Test the alias:

```
nslookup play.musicmania.in 127.0.0.1
# or
dig @127.0.0.1 play.musicmania.in A
```

**Expected result:** play.musicmania.in should appear as a CNAME pointing to www.musicmania.in. The final A-record lookup should return 192.168.1.50. A CNAME is an alias to another DNS name, not a direct IP address record.

**My test result:** _____
_____

## 4. DNS troubleshooting process

If musicmania.in does not resolve as expected, follow these checks in order:

**1. Check whether the DNS service is running:**

```
sudo systemctl status bind9
```

**2. Check the main configuration syntax:** No output normally means there are no syntax errors.

```
sudo named-checkconf
```

**3. Check the DNS zone file:** Look for spelling mistakes, missing dots, incorrect record types, or a wrong file path.

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

**4. Review the zone declaration and file path:**

```
sudo nano /etc/bind/named.conf.local
```

**5. Restart the DNS service or reload the zone:**

```
sudo systemctl restart bind9
sudo rndc reload musicmania.in
```

**6. Test DNS resolution directly against the local server:**

```
dig @127.0.0.1 www.musicmania.in A
dig @127.0.0.1 play.musicmania.in A
nslookup www.musicmania.in 127.0.0.1
```

**7. Check logs and whether DNS is listening on port 53:**

```
sudo journalctl -u bind9 --no-pager -n 30
sudo ss -lntup | grep ':53'
```

**Additional checks:** Confirm that the client is using the correct DNS server. Check firewall rules for DNS over UDP and TCP port 53. For tests from another machine, make sure BIND listens on the VM's network interface and allows queries from the lab network. Do not expose a test DNS server publicly.

## Troubleshooting record

| Check performed | What I found / action taken |
|---|---|
| Service status | _____<br/>_____ |
| Configuration / zone check | _____<br/>_____ |
| nslookup / dig result | _____<br/>_____ |
| Final result | _____<br/>_____ |

## Conclusion

In this practical, I learned how to install and configure a BIND DNS server, create a DNS zone, add A and CNAME records, verify DNS resolution using nslookup or dig, and troubleshoot common DNS problems.

**Student Signature:** _____ **Date:** _____